

A Project Report on

LIP READING USING DEEP LEARNING TECHNIQUES

submitted in partial fulfilment of the requirements for the award of the degree of

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Abstract

Visual Speech Recognition (VSR) or lip reading has been receiving extensive interest across areas of accessibility, security, and silent communication. While deep learning-based lip-reading techniques have demonstrated highly encouraging performance under well-controlled offline setups, deployment under real-world setups has always remained the major stumbling block due to problems related to latency, environmental variability, and speaker variability. This research explores the training of a deep learning lip reading system on the GRID database with each video being processed in fixed-length sequences of 25 frames and also achieving an accuracy of 97.2%. In addition, the system demonstrates robust performance in recognizing various speech patterns from different speakers. The system is then implemented in real-time using a webcam to predict structured sentences, achieving an accuracy of 50% in real-time predictions. The transition from offline training to real-time usage, however, is full of significant challenges such as reduced accuracy, increased misinterpretations, and performance instability due to real-world conditions. This report is a thorough analysis of such issues, with a specific focus on offline and real-time performance disparity. We also discuss potential enhancements, such as model generalization through multi-speaker training and real-time inference optimization. With the minimal previous work in real-time phrase prediction with webcams, this report is an important contribution to a very extensive but comparatively underexplored area of lip-reading technology.

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ABBREVIATIONS

AI	<i>Artificial Intelligence</i>
ML	<i>Machine Learning</i>
<i>Bi-LSTM</i>	<i>Bidirectional Long Short Term Memory</i>
<i>3D-CNN</i>	<i>3D Convolutional Neural Networks</i>
CTC	<i>Connectionist Temporal Classification</i>
WER	<i>Word Error Rate</i>

Chapter 1

Introduction

Lip reading plays a crucial role in scenarios where audio-based speech recognition is unreliable or unavailable. It has applications in assistive technology for the hearing-impaired, silent speech interfaces, and security systems. Advances in deep learning have significantly improved lip-reading accuracy, with models leveraging convolutional neural networks (CNNs) and recurrent architectures such as LSTMs and Transformers.

While lip-reading models perform well in offline settings, real-time implementation remains a complex challenge. Unlike pre-recorded videos where frames are processed in a structured sequence, real-time lip reading relies on continuous video input, making it susceptible to inconsistencies such as lighting variations, occlusions, and varying speaking speeds. Additionally, real-time inference demands efficient processing to maintain low latency without sacrificing accuracy.

Most research in deep-learning-based lip reading has focused on word-level or sentence-level prediction in offline settings, where models are trained on fixed datasets with well-aligned frames. However, transitioning these models to real-time prediction introduces several obstacles:

- Latency Constraints: Processing each video frame in real-time requires efficient inference, especially for complex deep-learning models.
- Environmental Variability: Changes in lighting, camera angles, and background noise impact performance.
- Speaker Adaptation Issues: Models trained on a single speaker may struggle to generalize to new speakers.
- Frame Synchronization: Real-time input often lacks precise frame alignment, affecting sequential predictions.

This research aims to:

- Train a deep learning model for phrase-level lip reading using the GRID dataset, where each video is segmented into 25-frame sequences.
- Implement the trained model in a real-time setting using a webcam to predict phrases as they are spoken.

This research is structured into two main parts: Part 1 describes the training process, including dataset selection, model architecture, and evaluation of offline performance. Part 2 focuses on real-time implementation, detailing system setup, challenges faced during live inference, and an analysis of performance limitations. Finally, we discuss possible improvements and future directions for achieving more reliable real-time lip reading.

Chapter 2

Literature Survey

Lip reading has been extensively studied in offline settings, where models process pre-recorded videos to predict spoken words or sentences. Several deep learning-based approaches have significantly improved lip-reading accuracy, leveraging convolutional and recurrent networks.

One of the earliest breakthroughs in sentence-level lip reading was LipNet, an end-to-end neural network that achieved 95.2% accuracy on the GRID dataset by utilizing spatiotemporal convolutions and recurrent networks, outperforming both word-level models and human lip readers [1]. Similarly, LipSyncNet employed a 3D-CNN with EfficientNetB0 and a Bi-LSTM backend, achieving 96.7% accuracy on GRID by effectively extracting visual features from lip movements in audio-challenged environments [2].

In addition, Transformer-based approaches have shown promising results. A 3D Convolutional Vision Transformer (3DCvT) achieved state-of-the-art performance on large-scale lip-reading datasets such as LRW and LRW-1000, demonstrating the potential of attention-based architectures in visual speech recognition [3]. Other works, such as the "Watch, Listen, Attend, and Spell" (WLAS) model, have explored audio-visual fusion

methods, outperforming previous models on benchmark datasets and even surpassing professional lip readers in some scenarios [4].

More recently, real-time lip-reading systems have begun to emerge. One study introduced a real-time automatic lipreading system for isolated English digits, achieving 93% accuracy for speaker-dependent recognition and 84% for speaker-independent recognition. This system leveraged a 14-point ASM lip model and incorporated both outer lip contours and inner mouth features to enhance performance in noisy environments [5].

Further advancements have been seen with LCANet, an end-to-end deep neural network architecture designed for machine lip reading. This model introduced a cascaded attention-CTC decoder alongside a stacked 3D-CNN, highway network, and bidirectional GRU encoder, demonstrating significant improvements in recognition accuracy [6]. Additionally, a neural network-based lip-reading system using visual cues represented by a limited number of visemes has been proposed, achieving remarkable performance improvements on the BBC LRS2 dataset by predicting full sentences from lip movements [7].

Most of these studies focus on controlled offline settings, where pre-segmented video clips are processed in structured environments. However, real-time lip reading, where predictions must be made continuously from live webcam input, remains a significantly underexplored challenge. Some efforts have been made to extend lip-reading models to profile views using multi-view datasets and curriculum learning approaches [8], but real-time phrase prediction has received little attention.

The GRID[9] dataset consists of 34 speakers, each uttering 1000 sentences, amounting to a total of 34,000 video clips. Each sentence follows a structured six-word format, ensuring a controlled vocabulary and grammatical consistency. The dataset was originally designed for audio-visual speech recognition and provides both video and audio recordings of speakers uttering phrases in a fixed frontal view with minimal background noise.

Despite advancements in lip-reading technologies, there is a notable lack of research focused on implementing real-time lip-reading systems using standard webcams in uncontrolled environments. Existing studies often concentrate on controlled datasets or specific applications, such as character recognition for individuals with speech impairments. This gap

underscores the need for research addressing the challenges of real-time lip-reading in more variable and practical settings.

In this work, we aim to bridge this gap by implementing a deep learning-based lip-reading model trained on the GRID dataset and evaluating its performance in a real-time setting using a webcam. Unlike previous studies, our focus is on phrase-level prediction in a real-time environment, where factors like latency, environmental variations, and synchronization challenges significantly impact accuracy.

Chapter 3

Limitation of Existing system

Despite significant advancements in the field of visual speech recognition, existing lip reading systems still face several challenges that limit their performance and real-world applicability. Some of the key limitations are as follows:

1. Dependency on Full-Length Video Input

Most traditional lip reading models are designed to process entire video sequences to generate predictions. This approach is not practical for real-time applications, where fixed-length frame processing is required to ensure low-latency and continuous predictions. The inability to handle streaming input restricts these models from being deployed in real-time scenarios such as video conferencing, security, or accessibility tool

2. Lack of Generalization to Real-Time Scenarios

Many models trained on datasets like GRID are optimized for offline performance, where temporal context and complete utterances are available. These models struggle to maintain accuracy when limited to short, fixed frame windows (e.g., 25 frames) as required in real-time systems.

3. High Computational Complexity

Deep learning-based lip reading systems, especially those involving complex architectures like 3D CNNs, Bi-LSTMs, and attention mechanisms, demand high computational power. This hinders their deployment on edge devices or low-power environments needed for real-time applications.

4. Limited Robustness to Speaker and Environmental Variability

Existing systems often perform well on constrained datasets like GRID, which involve limited speakers, consistent lighting, and frontal face views. However, their performance significantly drops when exposed to diverse speakers, accents, head poses, or background noise—conditions that are common in real-world, real-time applications.

5. Delayed or Inaccurate Predictions in Streamed Input

In real-time use cases, predictions must be made on the fly. Existing models typically require the full context of a sentence for accurate predictions. When restricted to short input clips (e.g., 1 second or 25 frames), their accuracy degrades due to insufficient temporal information.

6. Dependence on Controlled Environments

Most state-of-the-art systems have been evaluated on curated datasets with controlled lighting, frontal face angles, and consistent speaking speed. Such conditions are rarely replicated in the real world, where users may move, speak at varying speeds, or be recorded in dimly lit environments.

7. Inadequate Handling of Speaking Variability

Lip movements can differ drastically from one individual to another due to differences in: Accent

- Facial structure
- Lip thickness
- Speaking style Most models fail to generalize to unseen users without significant retraining.

Chapter 4

Problem Statement and Objectives

In a world that is becoming increasingly reliant on digital communication, there remains a critical need for technology that can bridge communication gaps. Real-time lip-reading technology has the potential to revolutionize how people with hearing impairments, individuals in noisy environments, or those who need to communicate discreetly can interact more effectively.

Problem Overview:

Despite advancements in speech recognition, understanding speech in challenging environments, such as noisy settings or for individuals with hearing disabilities, remains a significant problem. Current solutions often struggle with accurately interpreting speech in real-time, particularly when audio cues are absent or when multiple people are speaking simultaneously. Traditional speech recognition systems may fail when there is background noise, poor audio quality, or speech is garbled, leaving a communication gap.

Lip reading, or visual speech recognition, offers an innovative solution to this issue. It leverages visual cues movement of the lips, face, and sometimes gestures to understand what someone is saying, even without the use of audio. The challenge is to develop an accurate and scalable real-time lip-reading system that can interpret these visual cues in various lighting conditions, for different speakers.

Objectives:

- Propose a deep learning-based lip reading model architecture using components like 3D CNNs, Bi-LSTMs, and Attention mechanisms, optimized for real-time performance through hyperparameter tuning and learning rate scheduling.
- Implement and adapt the proposed model for real-time lip reading, ensuring efficient inference by processing fixed-length frame windows and integrating a real-time video processing pipeline.
- Evaluate the system's performance by comparing real-time predictions against ground truth transcriptions, focusing on both accuracy and latency metrics.

Chapter 5

Proposed System

5.1 Algorithm and Frameworks used

- **3D Convolutional Neural Networks (3D CNNs)**

3D CNNs are an extension of traditional 2D CNNs, designed to process sequences of images like video clips rather than single images. While 2D CNNs operate over height and width (spatial features), 3D CNNs also include the time dimension (depth). This means they can capture spatiotemporal features, i.e., both the motion across frames and the spatial structure within each frame.

In lip reading, this is very useful because it allows the model to understand how the lips move over time, not just how they look in one frame. A 3D CNN processes a short sequence of frames together (like your 25-frame input) and outputs feature representations that encode both space and time.

- **Bidirectional Long Short-Term Memory Networks (Bi-LSTMs)**

LSTMs are a type of Recurrent Neural Network (RNN) that can remember information over long sequences. Bi-LSTMs improve on regular LSTMs by processing data in both forward and backward directions. This is helpful when the full context of a sequence is important for

understanding like in lip reading, where knowing what came before and after a particular lip movement improves prediction accuracy.

For example, to guess a word correctly, the model might need to know the surrounding words or sounds, which Bi-LSTMs can provide because they “see” the entire sequence from both ends.

- **Attention Mechanism**

The attention mechanism helps models focus on the most relevant parts of an input sequence when making predictions. Instead of treating all frames or features equally, attention allows the model to weigh important frames more heavily for example, frames where the mouth is moving significantly.

This is useful in lip reading because not all frames contribute equally to a spoken word. Some may show silence or irrelevant motion, and attention helps the model skip those and focus on the meaningful visual cues.

- **Connectionist Temporal Classification (CTC) Loss**

CTC loss is used when the input (like 25 video frames) and output (text) don't align perfectly that is, we don't know which frame corresponds to which character. CTC solves this by allowing the model to learn the alignment on its own and produce variable-length outputs from fixed-length inputs.

In lip reading, this is especially useful because people speak at different speeds, and the same word can be spoken in fewer or more frames depending on the speaker.

- **FastAPI**

FastAPI is a modern, lightweight web framework in Python used for building APIs quickly. It's built on top of Starlette and Pydantic, and it supports asynchronous programming — making it fast and ideal for real-time applications.

In your project, FastAPI is used to expose the trained model through an API so it can receive frame inputs from a web app and return predictions efficiently.

- **WebRTC (Web Real-Time Communication)**

WebRTC is a technology that allows real-time audio, video, and data sharing directly between browsers. It's peer-to-peer and doesn't require external plugins.

In your lip reading system, WebRTC is used to capture video from the user's webcam in real time and stream it to the backend server, enabling the real-time component of your application.

- **WebSockets**

WebSockets allow persistent, two-way communication between a client and server. Unlike normal HTTP requests which are one-way, WebSockets enable the server to send data back to the client without the client having to request it every time.

In your system, WebSockets are used to send video frame predictions from the server to the frontend as soon as they're ready, reducing delay and supporting real-time feedback.

- **Word Error Rate (WER)**

Word Error Rate (WER) is a standard evaluation metric used to assess the accuracy of sequence-to-sequence models in speech and lip reading tasks. It measures how closely the predicted text output matches the ground truth by calculating the minimum number of word-level errors required to convert the prediction into the actual sentence.

WER is calculated using the formula:

$$\text{WER} = (S + D + I) / N$$

Where:

- ❖ S = Number of substitutions
- ❖ D = Number of deletions
- ❖ I = Number of insertions
- ❖ N = Total number of words in the ground truth

The resulting value represents the proportion of errors with respect to the total words in the reference. A **WER of 0.0** indicates a perfect prediction, whereas higher values indicate more significant errors.

In lip reading, WER provides an intuitive measure of how understandable the predicted sentence is. Since lip reading models may generate words with subtle visual cues, WER helps quantify real-world usability by penalizing incorrect, missing, or extra words in predictions.

5.2 Design Details

1. Overview of the Proposed System

This work presents the development of a real-time lip reading system capable of converting silent visual speech from live video into text in near real-time. The primary motivation stems from the observation that, although numerous approaches exist for offline lip reading, real-time implementation remains largely underexplored in existing literature.

A single end-to-end model was used for both offline and real-time scenarios. It was trained from scratch using fixed-length input sequences (25 frames), with no architectural modifications required during transition to real-time inference.

2. Model Architecture

The proposed model follows a sequential architecture consisting of the following components: 3D Convolutional Neural Network (3D CNN) → Bidirectional LSTM (Bi-LSTM) → Scaled Dot-Product Attention → Dense Layer → Connectionist Temporal Classification (CTC).

- **3D CNN:** Employed for extracting spatio-temporal features from input video sequences. It comprises multiple convolutional layers with increasing filter sizes (e.g., 64, 128, 256), ReLU activations, and max pooling for dimensionality reduction.
- **Bi-LSTM:** A single-layer bidirectional LSTM processes the temporal features across the 25-frame sequence, capturing both forward and backward temporal dependencies.
- **Attention Mechanism:** A **scaled dot-product attention** mechanism enhances temporal focus, allowing the model to attend to the most relevant frame-level features during decoding.
- **CTC Loss:** The use of CTC is justified by the absence of precise alignment between lip movements and the corresponding character-level text, making it well-suited for sequence-level supervision.

3. Real-time Adaptation Design

To enable real-time performance, the system implements a buffering mechanism based on a sliding queue:

- **Frame Window:** The model was trained using 25-frame sequences, which correspond to approximately one second of video at 25 frames per second. This same window size was retained during real-time operation for consistency.
- **Buffering Strategy:** A **live queue** accumulates incoming frames from the video stream. Once 25 frames are available, they are processed as a single batch for prediction.
- **Inference Latency:** The end-to-end processing time for each 25-frame window ranges between **250 ms to 500 ms**, depending on the system load and frame ingestion rate.

4. Frontend and Backend Interaction

The system architecture comprises a lightweight frontend interface and a FastAPI-based backend, designed for low-latency communication:

- **Frontend Stack:** Implemented using HTML, CSS, and JavaScript, with WebRTC facilitating real-time video capture and transmission from the user's camera.
- **Communication Protocol:**
 - **Frame Transmission:** Live video frames are transmitted from frontend to backend using WebRTC.
 - **Prediction Feedback:** The backend sends the predicted text and related updates back to the frontend via WebSockets.
- **Backend Framework:** The backend is built using FastAPI, which receives the incoming video frames (e.g., pre-processed as base64), performs real-time inference, and returns textual predictions over WebSocket.

5.3 Methodology

5.3.1 Training a Lip Reading Model

We use the GRID audiovisual dataset, a widely used benchmark dataset for visual speech recognition and lip reading[9]. The GRID dataset includes 34 speakers, and each of them speaks 1000 sentences, making a total of 34,000 video clips. A well-defined six-word structure is assigned to each sentence to possess a controlled vocabulary and grammatical structure. The dataset was created for audio-visual speech recognition and offers both video and audio recordings of speakers speaking phrases in a fixed frontal position with minimal background noise. The GRID dataset employs a specific set of sentence construction rules within six categories: command, colour, preposition, letter, digit, and adverb. Each sentence employs a specific six-word structure that imposes a list of 51 unique words, four commands, four colours, four prepositions, 25 letters, 10 digits, and four adverbs. They are employed simultaneously at random to construct numerous different sentences without changing the grammar

We are using a subset of the data for our experiment. We choose particularly five speakers and so we have 5000 videos in all to train and test.

We propose a structured pipeline in Fig. 5.1 that processes videos suitable for real-time into meaningful sequences while preserving crucial spatial and temporal information for accurate predictions. Our system follows a well-defined multi-stage approach: preprocessing, feature extraction, temporal modelling, and sequence decoding. Each stage plays a crucial role in ensuring that the model learns rich representations while maintaining efficiency in processing. The system is designed to process videos in chunks of 25 frames per sequence while ensuring that the final prediction is made at the video level, rather than at the chunk level.

Before feeding raw videos into the model, it is essential to apply a series of preprocessing steps to ensure consistency in input representation. The system first segments each input video into chunks of 25 frames, maintaining the temporal flow of lip movements. Since the mouth region carries the most relevant information for lip reading, each frame undergoes lip region cropping, where only the area surrounding the lips is retained.

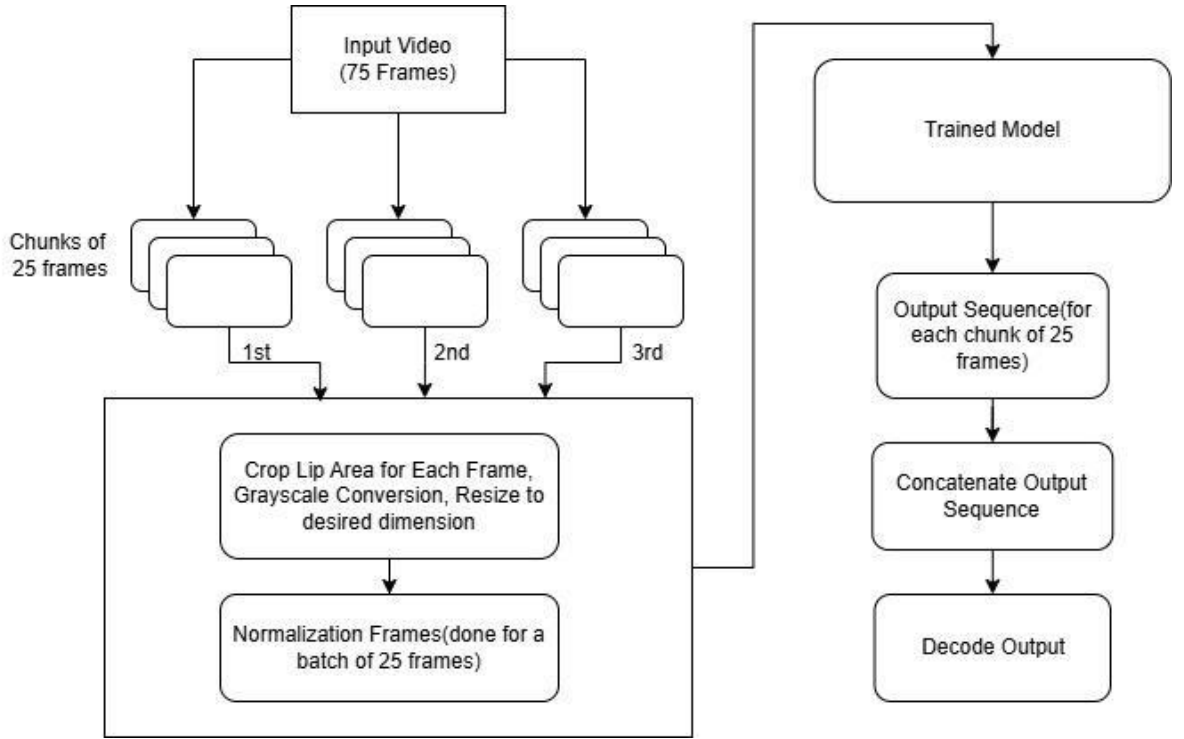


Fig. 5.1 Proposed System Design

This reduces unnecessary background noise and focuses the model on the critical visual information. To further optimize the input representation, the cropped frames are converted into grayscale, removing redundant colour information while preserving key contrast features that define lip movement. The grayscale frames are then resized to a fixed dimension of 140×40 pixels, ensuring uniformity across all inputs. Additionally, normalization is applied to scale pixel values, improving model stability and convergence during training. These pre-processed frames are then stacked sequentially into 3D tensors and passed into the trained model. The processed video chunks serve as input to the feature extraction module, which is based on 3D Convolutional Neural Networks (3D CNNs). Unlike traditional 2D CNNs, which operate on spatial dimensions alone, 3D CNNs are well-suited for video processing as they can capture both spatial and short-term temporal features simultaneously. This is particularly beneficial for lip reading, where motion patterns play a significant role in distinguishing phonemes and words.

Our model shown in Fig. 5.2 employs three 3D convolutional layers, each followed by an activation function (ReLU) and max pooling operations. The first convolutional layer extracts low-level features such as edges and texture variations in the lip region. As we progress deeper into the network, the subsequent convolutional layers capture higher-level abstractions,

including complex lip movements and phoneme-specific motion patterns. Max pooling operations are used to downsample the feature maps, reducing computational complexity while retaining the most informative regions.

By the end of this stage, the model produces a rich feature representation of the lip movements, structured as a temporal sequence of feature vectors. These extracted features are then passed into the temporal modelling component to capture long-term dependencies in speech articulation.

To model the sequential nature of lip movements, the extracted features are fed into a Bidirectional Long Short-Term Memory (Bi-LSTM) network. Traditional LSTMs are well-known for their ability to handle sequential data by learning dependencies across time steps. However, bidirectional LSTMs further enhance this capability by processing the sequence in both forward and backward directions. This is particularly beneficial for lip reading, as certain phonemes can be better recognized by considering both past and future contexts. A key design choice in our model is the use of stateful LSTMs, which ensures that the model maintains continuity across consecutive frame chunks. Instead of treating each 25-frame chunk independently, the LSTM states are preserved across chunks within the same video, allowing the model to leverage long-term dependencies. However, once the entire video has been processed, the LSTM states are reset before predicting the next video. This ensures that temporal continuity is maintained within a single video while preventing cross-video interference.

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 75, 46, 140, 1)	0	-
conv3d_3 (Conv3D)	(None, 75, 46, 140, 128)	3,584	input_layer_1[0][0]
activation_3 (Activation)	(None, 75, 46, 140, 128)	0	conv3d_3[0][0]
max_pooling3d_3 (MaxPooling3D)	(None, 75, 23, 70, 128)	0	activation_3[0][0]
conv3d_4 (Conv3D)	(None, 75, 23, 70, 256)	884,992	max_pooling3d_3[0][0]
activation_4 (Activation)	(None, 75, 23, 70, 256)	0	conv3d_4[0][0]
max_pooling3d_4 (MaxPooling3D)	(None, 75, 11, 35, 256)	0	activation_4[0][0]
conv3d_5 (Conv3D)	(None, 75, 11, 35, 75)	518,475	max_pooling3d_4[0][0]
activation_5 (Activation)	(None, 75, 11, 35, 75)	0	conv3d_5[0][0]
max_pooling3d_5 (MaxPooling3D)	(None, 75, 5, 17, 75)	0	activation_5[0][0]
time_distributed_1 (TimeDistributed)	(None, 75, 6375)	0	max_pooling3d_5[0][0]
bidirectional_2 (Bidirectional)	(None, 75, 256)	6,660,096	time_distributed_1[0]...
dropout_2 (Dropout)	(None, 75, 256)	0	bidirectional_2[0][0]
bidirectional_3 (Bidirectional)	(None, 75, 256)	394,240	dropout_2[0][0]
dropout_3 (Dropout)	(None, 75, 256)	0	bidirectional_3[0][0]
attention_1 (Attention)	(None, 75, 256)	1	dropout_3[0][0], dropout_3[0][0]
dense_1 (Dense)	(None, 75, 41)	10,537	attention_1[0][0]

Total params: 8,471,925 (32.32 MB)

Trainable params: 8,471,925 (32.32 MB)

Non-trainable params: 0 (0.00 B)

Fig5.2 Model Architecture

To prevent overfitting and improve generalization, dropout layers are incorporated after the Bi-LSTM layers. These layers randomly deactivate a fraction of neurons during training, forcing the model to learn more robust features.

One of the challenges in lip reading is that not all frames contribute equally to the final prediction. Some frames may carry more discriminative information, while others may be less relevant due to noise or occlusion. To address this, our model integrates an attention mechanism, which dynamically assigns varying importance weights to different frames. This

allows the model to focus more on the most informative parts of the sequence, leading to better accuracy and robustness. The attention-weighted feature vectors are then passed to a fully connected dense layer, which maps them to a character sequence output. This dense layer acts as the final decision-making unit, predicting the most likely character sequence based on the learned feature representations.

Once the character sequences of individual 25-frame segments are produced, the outputs are combined to produce the overall prediction of the whole video. This implies that while the model is processing smaller segments, the final output is specific to the whole speech. The text is decoded from the combined predictions to produce the final transcribed output for the video. This approach ensures that the model can process the lip movements well along with an uninterrupted flow from the frames. Using 3D CNNs for feature extraction, Bi-LSTMs for sequence modelling, attention for importance adjustment, and a fully connected layer for the final prediction produces an extremely efficient lip-reading system that can be used in real-time.

The learning timeline of the lip reading model has been planned in such a manner that it allows stable learning, fast convergence, and improved accuracy with time. We have used the Adam optimizer and have trained the model for 100 epochs. The entire learning process has been divided into two major phases: fixed learning rate based initial phase and adaptive learning rate scheduling method based second phase.

During the initial phase, the model was trained for 50 epochs using a fixed learning rate of 0.0001. This initial phase helped the model to learn fundamental spatial and temporal patterns of the data without sudden learning dynamics changes. Because a very high learning rate may introduce instability and hinder the model to learn patterns in an appropriate manner, this small but fixed learning rate ensured smooth optimization. During this phase, the model learned to learn informative lip movement features step by step and enhance its capacity to make sequence predictions. But applying a fixed learning rate for a long duration may cause slow convergence, and therefore the training approach should be modified to speed up learning while ensuring stability. In a bid to maximize the model performance after 50 epochs, the model was initialized with a dynamic learning rate scheduler.

Description	Epochs	Learning Rate	Training Loss
Stable Learning	1- 50	0.0001 (constant)	18.045
Warmup Phase	51 - 59	0.0001- 0.000515	9.334
Step Decay	60 - 100	0.000515- 0.000064	2.2406

Table 1 : Learning Rate Adjustments Over Epochs For Loss Progression

The scheduler used a two-stage model, beginning with a warmup stage followed by a step decay policy. The warmup stage of the model was run from epochs 51 to 59, where it gradually increased the learning rate by multiplying it by a factor of 1.2 at each epoch. Throughout this stage, the model was able to escape local minima and converge to more distinctive feature representations. By briefly increasing the learning rate, the optimizer was able to explore a better set of weights, leading to a corresponding improvement in the model’s generalization performance. After the model had finished the warmup stage, the learning rate was used to a step decay schedule where the learning rate was halved every 20 epochs. This led to weight updates being carried out in a more controlled fashion in the region of convergence, particularly avoiding overshooting of the optimal point. Decreasing the learning rate in subsequent epochs enabled the model to refine its generated representations and avoided excessive oscillation in the value of the losses.

Apart from it having an adaptive learning rate, the model was also trained with the Connectionist Temporal Classification (CTC) loss function. This is because CTC loss is particularly well-suited to be employed for tasks like lip reading, where input frame to output sequence alignment is not available. Most conventional sequence-to-sequence models need strict frame-to-label alignment, but with CTC, learning is possible without strict alignments.

It is thus best suited for processing lip movement, where the same speaker can utter a word differently, with variations in duration. CTC loss predicts character sequences with blank labels, which are used to fill in silence areas or the transitional area between phonemes. From the minimization of CTC loss, the model can learn to map complex lip movement patterns to the corresponding text outputs step by step.

5.3.2 Implementing the Model for Real-Time Prediction

The real-time inference system is designed to function efficiently on a machine without a GPU, leveraging a multithreaded architecture to handle continuous frame transmission and processing. The frontend captures video frames using a webcam or an external camera and sends them sequentially to the backend, where a queue is maintained to store incoming frames. Once the queue accumulates 25 frames, the model initiates the prediction process, treating the collected frames as a single input sequence.

During inference, the system ensures that while predictions are being computed, new frames continue to arrive without disruption. This parallel execution prevents bottlenecks and maintains a near-real-time flow. As soon as the prediction is generated, the system removes the processed frames from the queue, allowing space for the next sequence while simultaneously sending the predicted text back to the frontend using WebSockets. This ensures low-latency communication and near-instantaneous feedback.

The cycle repeats seamlessly until the user decides to stop the prediction process via the user interface. Upon receiving the stop signal, the frontend ceases frame transmission to the backend, allowing any remaining frames in the queue to be processed. At this stage, the Bi-LSTM states, which retain temporal dependencies for better sequence prediction, are explicitly reset. This is necessary because the model operates in a stateful manner, and retaining past states beyond a session could introduce inconsistencies in future predictions. Resetting ensures that each new session starts with a fresh state, preventing interference from previous inputs.

This structured yet efficient approach allows real-time lip reading to function smoothly even on hardware without dedicated GPU acceleration. By carefully managing frame queues, leveraging multithreading, and ensuring a continuous flow of data, the system strikes a balance between inference speed and processing efficiency, making real-time lip reading feasible in a constrained computing environment. The Implementation UI is shown in Fig. 5.4.

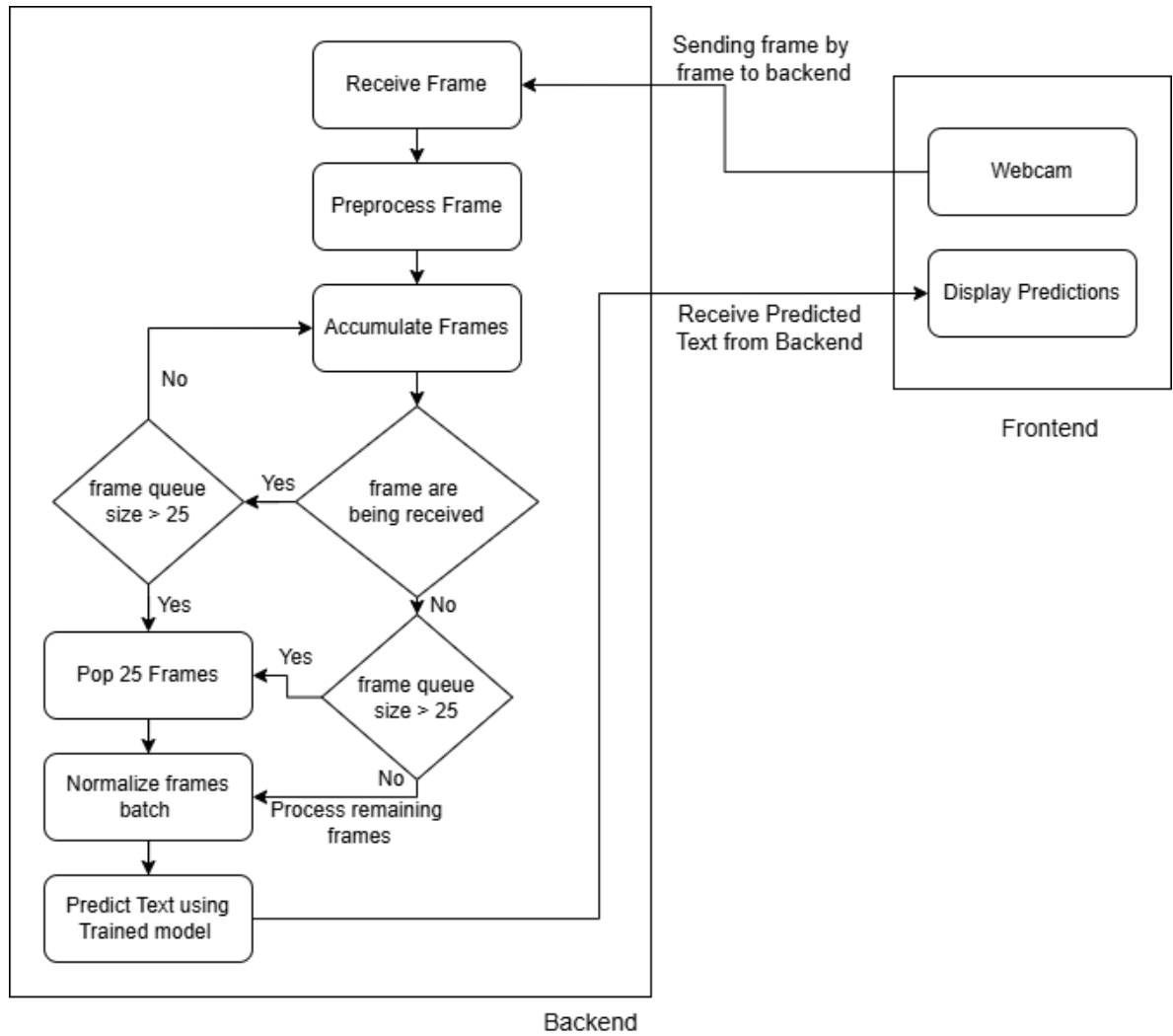


Fig. 5.3 Real-Time Lip Reading Inference Pipeline

This model demonstrated in Fig. 5.3 was tested on a real time system to evaluate its performance in predicting phrases as spoken. The system was deployed as a live video processor via webcam, capturing frames at 30 FPS and providing them to the trained model in batches of 25 frames. Each batch's predictions were then combined to generate the final phrase for the whole video. This real-time setup was employed to evaluate the performance of the model in accurately transcribing lip movements. Despite the official training paradigm and model optimizations, the hoped-for results could not be achieved in real-time inference.

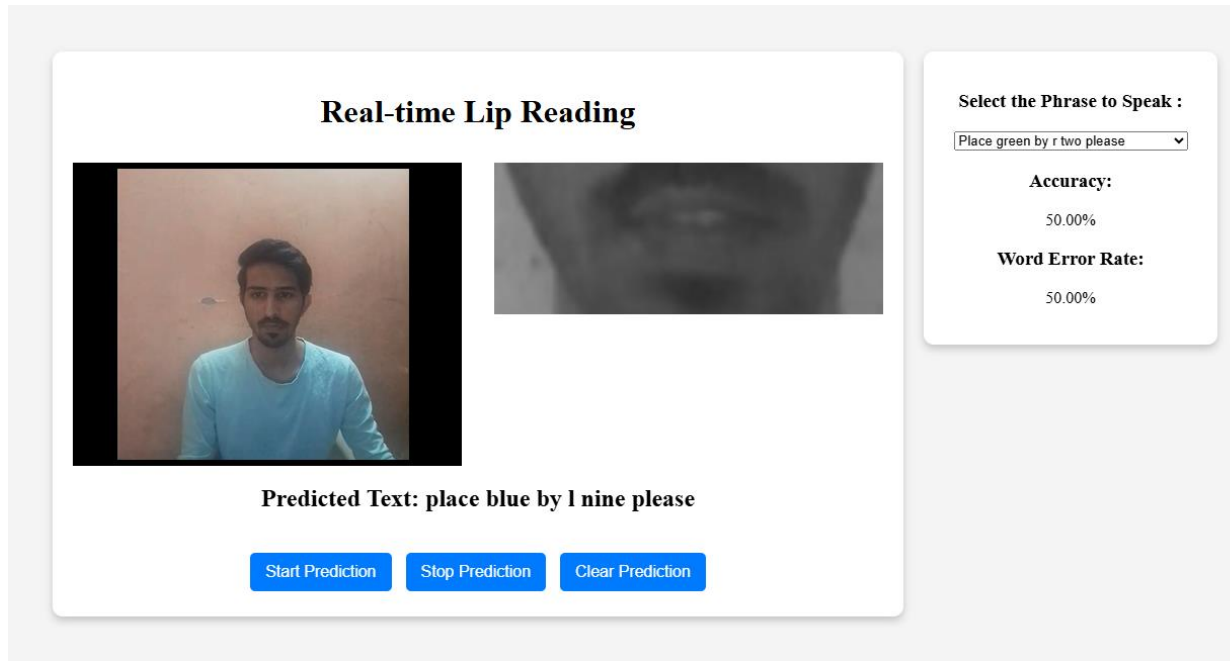


Fig. 5.4 UI Implementation for Real-Time Lip Reading

● Working of the Implementation

The real-time lip reading interface has been designed to allow users to interactively test the system by speaking predefined phrases and observing the model's predictions and accuracy in real time. The user interface is developed using HTML, CSS, and JavaScript for the frontend, with the backend built on FastAPI. WebRTC is used for real-time video capture, and WebSockets are employed for two-way communication between the frontend and backend, facilitating both the transmission of pre-processed frames and the display of predictions.

The process begins when the user clicks the "Start Prediction" button. This action initiates real-time video streaming from the user's webcam through WebRTC. As the stream progresses, the system continuously captures frames and appends them to a buffer (queue) on the backend. Once a sufficient number of frames (i.e., 25 frames, which roughly corresponds to 0.5–1 second of video) are accumulated, the oldest frames are popped from the queue and passed to the deep learning model for prediction.

The model pipeline follows the architecture: 3D Convolutional Neural Network (3D CNN) → Bidirectional LSTM (Bi-LSTM) → Scaled Dot-Product Attention → Dense Layer → Connectionist Temporal Classification (CTC) Decoder. Before feeding the frames into this model, they undergo preprocessing which includes grayscale conversion and lip region cropping using Mediapipe. These pre-processed frames are also displayed on the interface in

real time (as shown in the right sub-window of the main video feed), giving the user insight into what the model is actually processing.

While the prediction stream is active, the user is expected to speak the phrase selected from the dropdown menu on the right-hand side. The dropdown contains test phrases derived from the GRID dataset vocabulary. Once the phrase has been spoken, the user must click "Stop Prediction", which stops the frame capture and initiates evaluation. The system then compares the predicted text with the reference text (selected from the dropdown) to compute the Word Error Rate (WER) and accuracy.

The predicted sentence, along with the corresponding accuracy and WER, is then displayed on the interface. The prediction result allows users to assess how well the system understood the spoken input in real time. The "Clear Prediction" button allows users to reset the interface for a new trial.

This real-time implementation is an important component of the project, as it demonstrates the application of the trained model outside the offline evaluation context, in a realistic scenario where input timing, user behaviour, and external conditions such as lighting and background noise may vary. Moreover, this interface enables rapid experimentation and user feedback, making it a valuable tool for further refinements.

The model was not able to generate significant predictions, and even after applying edit distance correction, the output remained predominantly inaccurate with a best-case accuracy of 50% for some phrases. The offline and real-time performance difference reveals deep problems with deploying lip-reading systems into real, dynamic settings. The problems encountered with real-time inference, as well as areas of improvement, are discussed below.

- **Challenges Encountered in Real-Time Execution**

Real-time deployment of the lip-reading model proved to have some issues as can be observed that may not exist in the offline scenario.

- **Latency:** While WebRTC minimized the delay in frame delivery, model inference was still plagued by a perceptible lag. Since the system only predicts when 25 frames are queued, any delay in processing after the system waiting time resulted in a delay in providing real-time feedback. The delay was even more apparent when consecutive predictions had to be processed sequentially.

- **Environmental Variability:** Although offline controlled training environments offer consistent lighting, changing camera angles, and limited occlusions, real environments offer changing lighting, changing camera angles, and sporadic occlusions. Unstable lighting sometimes changed lip region contrast, which made it difficult for the model to acquire discriminative features. Similarly, slight changes in camera position affected the observability of significant lip movements, decreasing prediction accuracy.
- **Speaker Adaptation:** The system was trained on a formal data set in the first place, but real-world use introduced natural variations in lip movement patterns, pronunciation and speech rates. Even though training data from five speakers helped with generalization, the system continued to have poor performance with new users, indicating further dataset expansion and domain adaptation techniques.
- **Frame synchronization:** The Model works with constant 25-frame blocks, yet during real-time inference, the arriving frames don't necessarily stick to the order expected. Tiny variations in the frame arrival time or random loss of frames led to incorrect phrase segmentation, and prediction errors resulted.
- **Speaking Speed Mismatch:** Mismatch between training Fig. 6. Challenges encountered in Real-time Lip Reading data and actual scenarios regarding the speed of speaking. Speakers in GRID data speak at a uniform pace, whereas in actual scenarios, users speak either faster or slower than the dataset's average speed. Because the model anticipates lip movements to adhere to the patterns learned from training, speed of speaking differences can result in inaccurate predictions. The model might skip important frames when speech is faster, whereas slower speech might result in redundant repetition or incomplete sentence formation.
- **Prediction Control:** A challenge unique to this system was when a user requested the stop prediction action before the queue built up to 25 frames. Because the model was trained on sequences of precisely 25 frames, to predict anything on a sequence shorter than this led to unstable outputs. It was an issue because the model could not dynamically adjust to variable input lengths, leading to incomplete or incorrect phrase predictions when a premature stop was requested.

In addition, computational constraints were a challenge since the system was running without a GPU. Deep learning networks, especially those with 3D convolutions and recurrent networks, are typically optimized for hardware acceleration. CPU inference introduced processing bottlenecks, which, while mitigated by multithreading, contributed to system-level latency.

● **Potential Improvements for Real-Time Lip Reading**

To improve the performance and stability of real-time lip reading, various modifications can be made in various components of the system. The prominent challenges such as latency, environmental variation, frame synchronization, and speaker adaptation need specific solutions to fill the gap between offline and real-time performance.

- Reducing latency and computation bottlenecks are among the most critical points of improvement. While hardware accelerators such as GPUs or special AI chips can speed up inference significantly, model processing time is the root cause of latency. Model quantization and pruning [10] methods can reduce the computational burden by lowering the precision of model parameters or pruning redundant weights to expedite inference without losing accuracy. Other optimizations such as model compilation using TensorRT or executing it on edge-friendly frameworks can enhance real-time responsiveness.
- Frame synchronization problems are another vital enhancement. Because the model is operating on fixed 25 frame blocks, any live input synchronization discrepancy can lead to partial or wrong predictions. Frame interpolation methods [11] can be incorporated to make sure that in missing frames, a more fluid input stream is maintained.
- For handling variability in the environment, more pre-processing can be beneficial. Integration of real-time face tracking and lip area stabilization [12] can assist in minimizing inconsistencies due to camera motion and varying illumination. Adaptive contrast enhancement [13] can provide consistency in lip visibility across varying conditions.
- Speaker adaptation can be boosted through augmented training data and applying domain adaptation techniques [14]. Training on a larger, more heterogeneous set of speakers can make the model better generalize to unseen speakers. Fine-tuning on actual real-time user data of users conversing with the system can also make it more adaptable. Adaptive normalization techniques or few-shot learning can also make the model adapt better to different speaking styles.
- Another important improvement is handling cases where prediction is triggered before 25 frames are accumulated. Another strategy could be to modify the model to handle variable-length sequences so that it can make reasonable predictions even for incomplete sequences of less than 25 frames. This can be achieved by sequence padding,

attention-based methods that learn to handle short sequences, or a fallback method where the system discards incomplete input rather than making suspicious predictions.

- To address the challenge the disparity of speaking pace in training and real conditions poses, Dynamic Time Warping (DTW) [15] or Adaptive Temporal Alignment techniques[16] can be explored. The methods allow the model to account for varying speaking paces by more adaptively aligning real-time input against learned patterns. Data training augmentation [17] for the model, synthetically speeding up or slowing down training videos to make the model encounter a wider range of speaking paces, is another potential method.
- Lastly, exploring other models such as Transformer-based models or hybrid models, ensemble learning, combining convolutional and self-supervised learning approaches may enhance robustness. Transformers, as capable of processing long-range dependencies, may offer enhanced phrase-level prediction, and self-supervised learning may enable the model to be trained on learning more about regular real-world differences without requiring large amounts of labelled data

By integrating these improvements, the real-time lip-reading system can be made more accurate, more robust, and more responsive, closer to real-world deployment in real applications.

Chapter 6

Experimental Setup

6.1 Hardware Setup

- CPU: Intel Core i3 10th Gen
- Ram: 16 GB
- Cloud GPU for training model: NVIDIA Tesla P100 from Kaggle.

6.2 Software Setup

Programming Language:

- Python 3.11.5: Used as the core language for implementing, training, and deploying the deep learning lip reading model.

Deep Learning Frameworks & Libraries:

- TensorFlow 2.16.1 : Framework used to build, train, and evaluate the deep learning architecture for lip reading.
- OpenCV : Used for video frame extraction and basic image preprocessing operations.
- NumPy : Utilized for numerical operations and array manipulations throughout data preprocessing and model input handling.

Dataset Handling & Processing:

- GRID Dataset : Benchmark dataset consisting of audio-visual speech videos, used for training and evaluating the model.
- FFmpeg : Command-line tool used to convert and preprocess video files from the GRID dataset.
- Mediapipe : Used to detect and crop the mouth region from each video frame for efficient lip-focused input.

Development Environment:

- Kaggle Notebook: Cloud platform with GPU support, used for training the offline lip reading model.
- VS Code: Local IDE used for real-time system development and testing of the deployed model.

Backend:

- FastAPI: Lightweight web framework used to serve the trained model and handle client-server communication.
- WebRTC: Enables real-time video stream capture from the client browser.
- WebSocket: Maintains a live connection between frontend and backend for continuous frame and text prediction transfer.

Frontend:

- HTML, CSS, JavaScript: Used to build a simple interface that captures webcam input and displays real-time text predictions.

Chapter 7

Results and Discussion

To evaluate the performance of our proposed lip reading system, we performed both quantitative and qualitative assessments in offline and real-time settings.

Offline Evaluation

In the offline environment, the model was evaluated on a held-out test set comprising 1000 videos (20% of the dataset). The performance metric used was Word Error Rate (WER). The model achieved a WER of 2.81%, translating to an accuracy of 97.2%. This is on par with or better than several state-of-the-art approaches such as LipNet and LipSyncNet on the same GRID dataset. A breakdown of prediction results is as follows in table 2:

Table 2. Offline Evaluation Metrics for Lip Reading Model on GRID Dataset

Metric	Value
WER	2.81%
Accuracy	97.2%

Sample prediction examples in offline mode:

Table 3. Examples of Ground Truth vs Predicted Sentences in Offline Test

Ground Truth	Predicted Output
“Place green at h seven please”	“Place green at h seven please”
“Lay red in D three now”	“Lay red in D three now”
“Bin blue with A four again ”	“Bin blue with A four again”

As seen in table 3, the model performs almost perfectly when tested in offline mode, highlighting the strength of the model architecture.

Real-Time Evaluation

Real-time testing introduced new challenges. The same model was deployed through a webcam-based interface, with predictions made on-the-fly. Here, accuracy significantly dropped to approximately 50%, with variation depending on:

- Lighting conditions
- Camera quality
- Speaking pace
- Frame drop/inconsistencies

Table 4. Real-Time Lip Reading Predictions and Accuracy

Test Phrase	Real-Time Prediction	Accuracy (%)
"Place blue in D four soon"	"Place blue in D fo soon"	83.3%
"Place blue in D four soon"	"Bin red with three now"	66.6%
"Lay green at Z five again"	"green at five again"	50.0%
"Set white with A one soon"	"white A one"	33.3%
"Place red in C two now"	"Place red in wo now"	50.0%

The differences in prediction accuracy under real-time conditions are reflected in Table 4, which contains examples of actual predictions made during live webcam testing.

Comparative Analysis with Literature

Table 5. Comparison of Accuracy in Offline and Real-Time Settings with Other Models

System	Dataset Used	Accuracy (Offline)	Accuracy (Real-Time)
LipNet[1]	GRID	95.2%	Not Supported
LipSyncNet[2]	GRID	96.7%	Not Supported
Our Model	GRID	97.2%	50% (Avg)

As shown in table 5, our system shows comparable or better offline performance than prior models but struggles in real-time conditions, a trend noted as a common challenge in existing literature.

Limitations Noticed During Evaluation

1. Accuracy drop in real-time was consistently noticeable.
2. Inference latency was dependent on frame transmission and backend processing.
3. Environmental variation (e.g., lighting, face angle) heavily influenced model output.
4. Short prediction segments (<25 frames) resulted in unstable outputs.

Key Takeaways

- The system is highly capable in controlled (offline) settings.
- Real-time performance needs improvement in robustness, latency, and adaptability.
- External conditions must be accounted for in any real-world deployment.

Implications

The creation of an effective real-time lip-reading system has high potential in various applications. For the deaf, it could be an effective communication aid that translates visual speech signals into text or synthesized speech. In noisy environments where audio signals are masked, lip-reading technology could be used to enhance speech recognition systems, enhancing human-computer interaction. For security and surveillance purposes, the ability to read speech without hearing it can be used for the benefit of protecting privacy but can still be made operational. Real Time Lip-reading technology could enhance biometric security systems through the provision of an additional factor of authentication. The examination of

the unique lip movement patterns over certain sentences could be used as a biometric signature, enhancing the security systems.

Chapter 8

Conclusion and Future Work

In this study, we developed and evaluated a deep learning-based real-time lip-reading system capable of predicting spoken phrases from silent video input. Our approach integrated a 3D CNN for spatial and temporal feature extraction, an attention mechanism for enhanced focus on relevant features, and a Bi-LSTM backend for sequential modeling. The model was trained using the GRID dataset, employing a structured training strategy with an initial constant learning rate followed by a dynamic learning rate scheduler to improve convergence. The performance evaluation, conducted using Word Error Rate (WER), demonstrated the effectiveness of the model, achieving a WER of 0.0281, which translates to an approximate accuracy of 97.2%.

Throughout the development of this real-time system, we encountered several challenges, particularly in ensuring efficient processing speed while maintaining prediction accuracy. One of the primary difficulties was handling real-time inference without excessive computational overhead. The use of stateful LSTMs with states resetting after each video rather than each chunk was one of the key optimizations to address sequential dependencies efficiently. Additionally, processing frames in chunks of 25, coupled with careful pre-processing steps such as lip region cropping, grayscale conversion, normalization, and resizing, ensured that the input data was structured optimally for the model.

Despite the challenges, this research contributes to advancing real-time lip reading, highlighting the potential of deep learning-based approaches for silent speech recognition. With continued improvements in model architecture, training strategies, and optimization techniques, real-time lip-reading systems can become valuable tools in applications ranging from accessibility for the hearing impaired to security and silent communication technologies.

Future work could include addressing these limitations to enhance the system's efficacy:

- **Large and Multifaceted Datasets:** Scaling training to large vocabularies and spontaneous speech can allow the model to generalize better to real-world situations.
- **Improved Hardware Utilization:** Using GPUs or hardware accelerators can reduce computational bottlenecks, making inference faster and real-time interaction more fluid.
- **Pre-processing Robustness Techniques:** Pre-processing techniques with adaptability, i.e., real-time tracking of faces, Data Augmentation can enhance system robustness to changes in the environment.
- **Speaker Adaptation Processes:** Domain adaptation and transfer learning processes can enable the model to acquire speaker-dependent features and hence improve performance on different groups of users

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Appendix

A. GRID Dataset Sample Sentences

Each sentence in the GRID dataset follows a below six-word structure:

[Command] [Color] [Preposition] [Letter] [Digit] [Adverb]

Examples:

- "Place blue at D four now"
- "Set white with G three again"

Vocabulary Breakdown (51 words):

- **Commands:** place, set, lay, bin
- **Colors:** blue, red, green, white
- **Prepositions:** at, in, with, by
- **Letters:** A to Z (25 letters)
- **Digits:** 0 to 9 (10 digits)
- **Adverbs:** again, now, please, soon

B. Hyperparameters Used

Batch Size : 2

Epochs : 100

Initial Learning Rate : 0.0001

Optimizer : Adam

Scheduler Phase 1 : Constant LR (50 epochs)

Scheduler Phase 2 : Warmup + Step Decay

Dropout Rate (BiLSTM) : 0.3

Frame Window Size : 25 frames

Frame Size : 140×40 pixels

Loss Function : Connectionist Temporal Classification (CTC)

C. Preprocessing Steps

1. Lip Region Cropping using MediaPipe.
2. Grayscale Conversion to reduce redundant color info.
3. Resizing each frame to 140×40 pixels.
4. Normalization (pixel values scaled between 0–1).
5. Stacking frames into a 3D tensor of shape (25, 140, 140,1) for model input.

D. Library & Tool Versions

Python : 3.11.5

TensorFlow : 2.16.1

OpenCV : 4.11.0.86

FastAPI : 0.115.8

Mediapipe : 0.10.5

ffmpeg : 4.3.1

WebSockets : 14.2

NumPy : 1.26.4

Publication

The Research Paper is under scrutiny in ACROSET 2025 conference.

Submission Summary

Conference Name	IEEE International Conference on Advances in Computing Research on Science Engineering and Technology (ACROSET 2025)
Paper ID	443
Paper Title	Phrase Prediction for Real-Time Visual Speech Recognition
Abstract	Visual Speech Recognition (VSR) or lip reading has been receiving extensive interest across areas of accessibility, security, and silent communication. While deep learning-based lip-reading techniques have demonstrated highly encouraging performance under well-controlled offline setups, deployment under real-world setups has always remained the major stumbling block due to problems related to latency, environmental variability, and speaker variability. This research explores the training of a deep learning lip reading system on the GRID database with each video being processed in fixed-length sequences of 25 frames and also achieving an accuracy of 97.2%. The system is then implemented in real-time using a webcam to predict structured sentences. The transition from offline training to real-time usage, however, is full of significant challenges such as reduced accuracy, increased misinterpretations, and performance instability due to real-world conditions. This paper is a thorough analysis of such issues, with a specific focus on offline and real-time performance disparity. We also discuss potential enhancements, such as model generalization through multi-speaker training and real-time inference optimization. With the minimal previous work in real-time phrase prediction with webcams, this paper is an important contribution to a very extensive but comparatively lesser-studied area of lip reading technology.
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Primary Subject Area	Artificial Intelligence and Machine Learning
Submission Files	RLR -Manuscript.pdf (519.9 Kb, 4/22/2025, 6:49:02 PM)

Following is the research paper that is under scrutiny -

Phrase Prediction for Real-Time Visual Speech Recognition

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Index Terms—Deep Learning, Computer Vision, Visual Speech Recognition, Real-Time Lip Reading, Challenges in Real-Time Lip Reading

I. INTRODUCTION

Lip reading is especially relevant in situations where audio-based speech recognition is unavailable or not ensured. It is applied in hearing impairment assistive technology, silent speech interfaces, and security. Lip-reading has been greatly improved by deep learning with models utilizing convolutional neural networks (CNNs) and recurrent models like Long Short Term Memory LSTMs and Transformers.

Although offline lip-reading models perform satisfactorily, real-time deployment is a difficult task. In contrast to pre-recorded video where the frames are processed in a predeter-

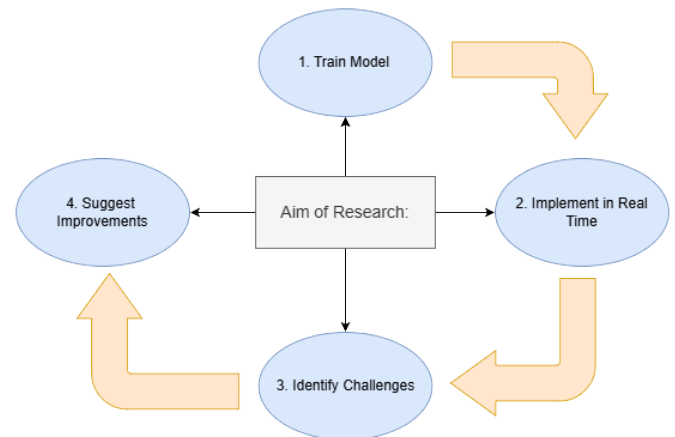


Fig. 1. Research Aim – Overview of the key steps involved in the study.

mined order, real-time lip reading is achieved by streaming video input and hence is prone to inconsistencies such as variations in lighting, occlusions, and differences in speaking paces. Moreover, real-time inference requires fast processing to ensure low latency without compromising accuracy. Most deep learning-based lip reading research has been sentence-level or word-level prediction [20] in offline settings, with models being trained on pre-fixed data with frames that are well-aligned. But it has some challenges to transfer the models to real-time prediction.

Fig. 1 shows the goal of this work where a deep model is trained for lip reading on a phrase level, employed online, and issues which arise later are identified along with suggestions to make it work even better. This paper is split into two sections: Part 1 describes training process, i.e., dataset selection, model architecture, and offline testing. Part 2 discusses implementation in real time, system architecture, live inference issues

faced, and performance limit analysis. We end with discussing possible areas of improvement and areas one could consider moving to in order to accomplish even more precise real-time lip reading.

II. RELATED WORK

Lip-reading has been researched for a long time on offline environments with pre-recorded videos, where the models learn words or sentences uttered. Deep-learning-based techniques have significantly enhanced lip-reading accuracy from the application of convolutional networks and recurrent networks.

One of the first sentence-level lip reading milestones was LipNet, an end-to-end neural network, which achieved 95.2% accuracy on GRID with spatiotemporal convolutions and recurrent networks and outperformed word-level models and human lip readers [1]. Similarly, LipSyncNet employed a 3D-CNN with EfficientNetB0 and a Bi-LSTM backend and achieved 96.7% accuracy on GRID because of its ability to extract visual features from lip motion in audio-overloaded conditions [2].

In addition, Transformer-based approaches have also reported promising results. 3D Convolutional Vision Transformer (3DCvT) achieved state-of-the-art performance on large datasets of lip-reading such as LRW and LRW-1000 and demonstrated that attention-based architectures can achieve strong performance on visual speech recognition [3]. Other studies such as the "Watch, Listen, Attend, and Spell" (WLAS) model have been experimenting with strategies of audio-visual fusion and performed well on benchmark sets and even superior to professional lip readers in some scenarios [4].

Recently, real-time lip-reading systems have started to appear. A paper proposed a real-time automatic lipreading system for isolated English digits, and it attained 93% speaker-dependent recognition accuracy and 84% in speaker-independent recognition. It used a 14-point ASM lip model and included both outer lip contours and inner mouth features to perform better in noisy conditions [5].

Greater strides have been achieved with LCANet, a deep neural network architecture for end-to-end machine lip reading. The model presented a cascaded attention-CTC decoder and stacked 3D-CNN, highway network, and bidirectional GRU encoder, demonstrating a dramatic recognition accuracy gain [6]. A visual cue-based lip-reading system based on a small viseme-count neural network was also presented, demonstrating spectacular performance gains over the BBC LRS2 dataset by predicting sentences from lip movements [7].

Most such work centers around controlled offline environments, with segmented video frames after segmenting being processed in organized environments. Real-time lip reading, in which predictions must be made in real time from live webcam feeds, is still a relatively underdeveloped task. While there have been attempts made to transfer the lip-reading models to multi-view dataset profiles and use curriculum learning methods [8], very little attention is given to the real-time phrasing prediction task.

Despite the progress in lip-reading technology, there is a glaring lack of studies focused on employing common webcams to apply real-time lip-reading systems in uncontrolled settings. Even though the majority of past contributions are focused on controlled data, there are limited contributions that elaborate on their adaptation to real-time, where performance greatly depends on latency, environment variations, and synchronization. That suggests that there is research to be done exploring the effectiveness of transferring lip-reading models to real-time, everyday application. In this paper, we try to fill this void by using a deep learning-based lip-reading model that has been trained using the GRID dataset and assessing how it performs on real-time basis using a webcam. Contrary to other research, we try to work within the scope of real-time phrase-level prediction and explore the issue that is seen when using a model trained from a controlled dataset to a changing environment.

III. METHODOLOGY

A. Training a Lip Reading Model

1) *Dataset*: We use the GRID audiovisual dataset for this research, a widely used benchmark dataset for visual speech recognition and lip reading[9]. The GRID dataset includes 34 speakers, and each of them speaks 1000 sentences, making a total of 34,000 video clips. A well-defined six-word structure is assigned to each sentence to possess a controlled vocabulary and grammatical structure. The dataset was created for audio-visual speech recognition and offers both video and audio recordings of speakers speaking phrases in a fixed frontal position with minimal background noise.

The GRID dataset employs a specific set of sentence-construction rules within six categories: command, color, preposition, letter, digit, and adverb. The specific pattern enables the construction of 64,000 unique sentences, including "place red at C two now," "set blue with H seven again," and "lay white at L three please." Each sentence employs a specific six-word structure that imposes a list of 51 unique words, four commands, four colors, four prepositions, 25 letters, 10 digits, and four adverbs. They are employed simultaneously at random to construct numerous different sentences without changing the grammar.

We are using a subset of the data for our experiment. We choose particularly five speakers and so we have 5000 videos in all to train and test.

2) *Data Pre-Processing*: Preprocessing is also an essential process in lip reading as it enhances the quality of input data to allow deep learning models to easily detect important visual features. Visual speech recognition is distinct from conventional speech recognition using acoustic signals and is dependent solely on small lip movements and hence demands a very clean input representation. Variability in light, speakers, sounds, and occluded faces introduces variability in the dataset and thus renders raw video frames of limited utility to train the models directly. Good preprocessing not only makes input data more consistent but also enables the model to function

well under various conditions, and consequently, improves lip-reading performance.

In this study, each video in the GRID dataset consists of 75 frames recorded at 25 frames per second. Our model is designed to perform well under real-time conditions, so we process the video in chunks of 25 frames at a time. This allows us to generate three alternative predictions for each video. This approach assists our system to generate results at consistent times, matching real-time requirements without disrupting the timing of lip movements.

To concentrate only on the essential visual information, we crop the lip region from every frame. Rather than considering the entire face, which includes additional details such as eyes, nose, and background, we trim the frames to input only the mouth region. This simplifies the processing and allows the model to learn lip movement. Because deep learning models perform more efficiently with constant input, we normalize the pixel values to keep brightness and contrast equal across all frames. This prevents changes in lighting from affecting what the model outputs.

We also transform the cropped lip area to grayscale. This removes extra color information because color isn't really helping in lip movement recognition. It lowers the number of input channels from three (RGB) to one, lowering processing time but retaining important motion information. We also resize the lip area to a fixed size of 140×46 pixels. This makes all frames have the same input size. This is important for deep learning models because changing input sizes may interfere with feature extraction and cause poor learning.

In addition to processing video frames, we also need to process text alignments. Each sentence in the GRID dataset also comes with a word-by-word transcription that must be transformed into model training form. The transcriptions are tokenized first, i.e., each word is separated into characters. The characters are then converted into numbers by a vocabulary set, thus enabling the model to process the text in a structured format. This processing of alignments allows the model to learn the correct mapping between lip movements and the text so that it can predict the words correctly. By carrying out these pre-processing steps, we have a normalized, information-dense representation of visual and text data both. This allows our model to learn the complex correlations between lip movements and speech well, so that it can be used in real-time lip reading.

3) *Model Architecture*: We introduce a systematic pipeline in Fig. 2 here that converts videos suitable for real-time processing into efficient sequences with spatial and time information for accurate predictions. We use a novel multi-step framework: preprocessing, feature extraction, temporal modeling, and sequence decoding. Each step is essential so that the model learns complex details without losing efficiency in processing. The system processes videos in sequences of 25 frames without losing the fact that the final prediction is made at the video level and not batch level.

Prior to the uploading of raw videos into the model, preparation processes must be done to normalize the input. The system

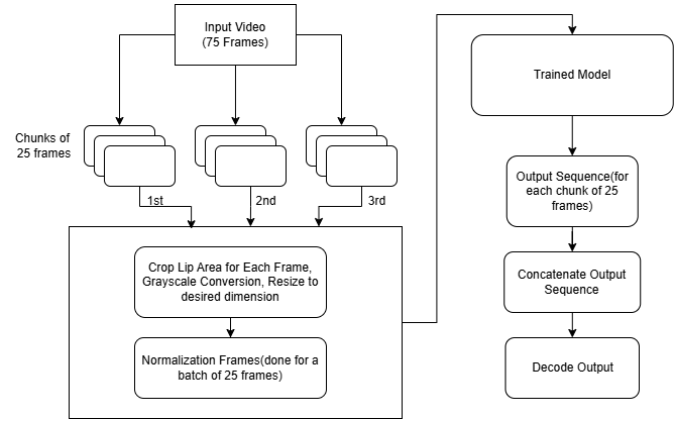


Fig. 2. Proposed System Design

then splits each of the videos into 25-frame segments with the lip movement's order maintained. As most of the information being utilized in lip reading is around the mouth region, all of the frames are cropped around the lips, keeping the lip edges intact. This eliminates further background noise and allows the model to concentrate on the correct visual information.

Other than that, the steps mentioned in data-preprocessing are done. Such processed frames are then merged in 3D forms and then provided to the trained model.

The preprocessed video clips are passed as inputs to the feature extraction module, which uses 3D Convolutional Neural Networks (3D CNNs). Whereas standard 2D CNNs can handle only space, 3D CNNs are better for video because they can handle space as well as short-term time features at the same time. This comes in handy in lip reading, where the pattern of movement is of utmost significance in identifying the sounds and words.

Our Fig. 3 architecture consists of three 3D convolutional layers. All are followed by ReLU activation function and max pooling operations. The first convolutional layer detects low-level features such as edges and texture variation in the lip region. As one goes deeper in the network, deeper convolutional layers detect higher-level features, e.g., complex lip motion and individual speech habits. Max pooling operations are employed to downsample the feature maps to simplify processing without discarding the most significant areas. Towards the end of this stage, the model produces a complete set of lip movement features, as a time series of feature vectors. The features are passed to the temporal modeling block to learn patterns of long-term speech construction. For sequentiality encoding of lip movement, the extracted features are passed through a Bidirectional Long Short-Term Memory (BiLSTM) network. Plain LSTMs are themselves already well-known to learn sequential data through learning between-frame relationships. But bidirectional LSTMs take this capability a step forward by passing the sequence to the network in both directions, forward and backward. This is very useful in lip reading because certain phonemes are better identified by including future and past context. One option in our model

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 75, 46, 140, 1)	0	-
conv3d_3 (Conv3D)	(None, 75, 46, 140, 128)	3,584	input_layer_1[0][0]
activation_3 (Activation)	(None, 75, 46, 140, 128)	0	conv3d_3[0][0]
max_pooling3d_3 (MaxPooling3D)	(None, 75, 23, 70, 128)	0	activation_3[0][0]
conv3d_4 (Conv3D)	(None, 75, 23, 70, 256)	884,992	max_pooling3d_3[0][0]
activation_4 (Activation)	(None, 75, 23, 70, 256)	0	conv3d_4[0][0]
max_pooling3d_4 (MaxPooling3D)	(None, 75, 11, 35, 256)	0	activation_4[0][0]
conv3d_5 (Conv3D)	(None, 75, 11, 35, 75)	518,475	max_pooling3d_4[0][0]
activation_5 (Activation)	(None, 75, 11, 35, 75)	0	conv3d_5[0][0]
max_pooling3d_5 (MaxPooling3D)	(None, 75, 5, 17, 75)	0	activation_5[0][0]
time_distributed_1 (TimeDistributed)	(None, 75, 6375)	0	max_pooling3d_5[0][0]
bidirectional_2 (Bidirectional)	(None, 75, 256)	6,660,096	time_distributed_1[0]-
dropout_2 (Dropout)	(None, 75, 256)	0	bidirectional_2[0][0]
bidirectional_3 (Bidirectional)	(None, 75, 256)	394,240	dropout_2[0][0]
dropout_3 (Dropout)	(None, 75, 256)	0	bidirectional_3[0][0]
attention_1 (Attention)	(None, 75, 256)	1	dropout_3[0][0], dropout_3[0][0]
dense_1 (Dense)	(None, 75, 41)	10,537	attention_1[0][0]

Total params: 8,471,925 (32.32 MB)
Trainable params: 8,471,925 (32.32 MB)
Non-trainable params: 0 (0.00 B)

Fig. 3. Model Architecture

design is using stateful LSTMs such that the model is capable of carrying continuity between chunks of successive frames. Rather than classifying each 25-frame chunk independently, LSTM states are propagated between chunks of the same video so that the model is capable of using long-term relationships. However, after processing an entire video, LSTM states are reset before predicting the next video. This keeps temporal continuity within a video without impacting between-videos.

To prevent overfitting and improve generalization, dropout layers are inserted between the BiLSTM layers. The dropout layers drop out certain neurons randomly at training time and hence the model learns robust features. One of the issues with lip reading is that not all frames are as useful for the final prediction. Some frames contain more useful information, while others are less useful due to noise or obstruction. Our model bypasses this by including an attention mechanism that assigns varying weights to varying frames. This implies that the model can focus more on the most useful sections of the sequence, improving accuracy and reliability. The attention-weighted feature vectors are then fed into a fully connected dense layer, which projects them to a character sequence output. This dense layer makes the final prediction, predicting the most likely character sequence from the learned features. Once the character sequences of individual 25-frame segments are produced, the outputs are combined to produce the overall prediction of the whole video. This implies that while the model is processing smaller segments, the final output is specific to the whole speech. The text is decoded from the

combined predictions to produce the final transcribed output for the video. This approach ensures that the model can process the lip movements well along with an uninterrupted flow from the frames. Using 3D CNNs for feature extraction, BiLSTMs for sequence modeling, attention for importance adjustment, and a fully connected layer for the final prediction produces an extremely efficient lip-reading system that can be used in real-time.

4) *Training Strategy*: The learning timeline of the lip-reading model has been planned in such a manner that it allows stable learning, fast convergence, and improved accuracy with time. We have used the Adam optimizer and have trained the model for 100 epochs. The entire learning process has been divided into two major phases: fixed learning rate based initial phase and adaptive learning rate scheduling method based second phase.

TABLE I
LEARNING RATE ADJUSTMENTS OVER EPOCHS FOR LOSS PROGRESSION

Description	Epochs	Learning Rate	Loss
Stable Learning	1 - 50	0.0001 (constant)	18.045
Warmup Phase	51 - 59	0.0001 - 0.000515	9.334
Step Decay	60 - 100	0.000515 - 0.000064	2.2406

During the initial phase, the model was trained for 50 epochs using a fixed learning rate of 0.0001. This initial phase helped the model to learn fundamental spatial and temporal patterns of the data without sudden learning dynamics changes. Because a very high learning rate may introduce instability and hinder the model to learn patterns in an appropriate manner, this small but fixed learning rate ensured smooth optimization. During this phase, the model learned to learn informative lip movement features step by step and enhance its capacity to make sequence predictions. But applying a fixed learning rate for a long duration may cause slow convergence, and therefore the training approach should be modified to speed up learning while ensuring stability. In a bid to maximize the model performance after 50 epochs, the model was initialized with a dynamic learning rate scheduler. The scheduler used a two-stage model, beginning with a warmup stage followed by a step decay policy. The warmup stage of the model was run from epochs 51 to 59, where it gradually increased the learning rate by multiplying it by a factor of 1.2 at each epoch. Throughout this stage, the model was able to escape local minima and converge to more distinctive feature representations. By briefly increasing the learning rate, the optimizer was able to explore a better set of weights, leading to a corresponding improvement in the model's generalization performance. After the model had finished the warmup stage, the learning rate was used to a step decay schedule where the learning rate was halved every 20 epochs.

This led to weight updates being carried out in a more controlled fashion in the region of convergence, particularly avoiding overshooting of the optimal point. Decreasing the learning rate in subsequent epochs enabled the model to refine its generated representations and avoided excessive oscillation

in the value of the losses. Apart from it having an adaptive learning rate, the model was also trained with the Connectionist Temporal Classification (CTC) loss function. This is because CTC loss is particularly well-suited to be employed for tasks like lip reading, where input frame to output sequence alignment is not available. Most conventional sequence-to-sequence models need strict frame-to-label alignment, but with CTC, learning is possible without strict alignments. It is thus best suited for processing lip movement, where the same speaker can utter a word differently, with variations in duration. CTC loss predicts character sequences with blank labels, which are used to fill in silence areas or the transitional area between phonemes. From the minimization of CTC loss, the model can learn to map complex lip movement patterns to the corresponding text outputs step by step.

5) *Performance Evaluation*: For comparison of the model’s performance, word error rate (WER) was the primary measure. WER measures the difference between the model’s predicted text and reference transcription. It is calculated as the ratio of the total number of substitutions, deletions, and insertions to the number of words in the reference transcription. A lower WER measures a lower error rate in the model’s predictions and hence improved performance.

TABLE II
EXPERIMENTAL FINDINGS WITH THE PREVIOUS MODELS

Models	Dataset	Accuracy	WER
3D-2D-CNN-Bi-LSTM-HMM [10]	GRID	91.4%	8.6%
LipNet [1]	GRID	95.2%	4.8%
LipSyncNet [2]	GRID	96.4%	3.6%
WAS [4]	GRID	97%	3.0%
Our Model	GRID	97.2%	2.8%

WER of our system is 0.0281, i.e., approximately 2.81% of the words predicted are incorrect. This is also equivalent to 97.2% word recognition rate, i.e., the system predicts approximately 97.2% of the words correctly and predicts them wrongly in the rest.

B. Implementing the Model for Real-Time Prediction

1) *System Setup for Real-Time Inference*: Real-time inference system is designed to be efficient on a machine without a GPU, employing a multithreaded structure to handle continuous frame transmission and computation. The frontend acquires video frames from a webcam or an external camera and passes them sequentially to the backend, where the incoming frames are buffered by a queue. When there are 25 frames in the queue, the model begins the prediction process, considering the collected frames as a single input sequence.

At inference, the system is tasked with ensuring that as predictions are being calculated, new frames keep streaming in without interruption. Parallel processing keeps the bottlenecks at bay while maintaining the stream near real-time. The system empties the processed frames from the queue immediately after prediction to make space for the next batch and returns the predicted text to the frontend via WebSockets in the pro-

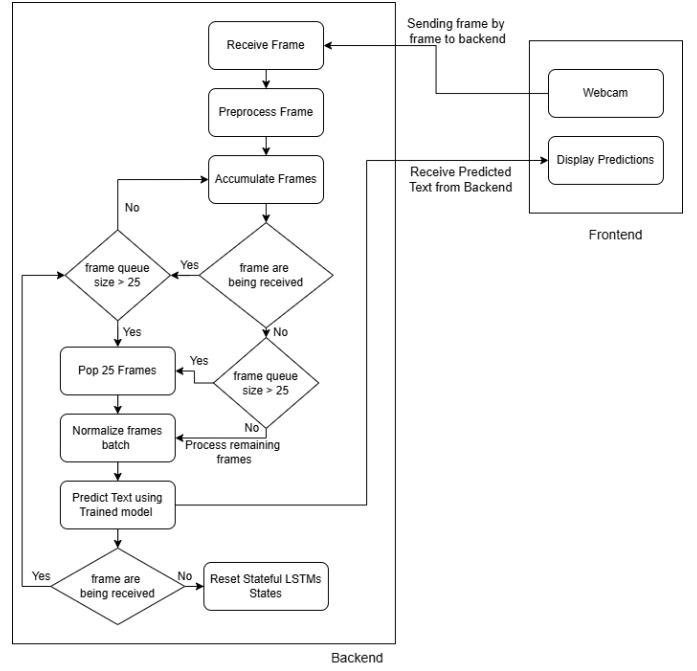


Fig. 4. Real-Time Lip Reading Inference Pipeline

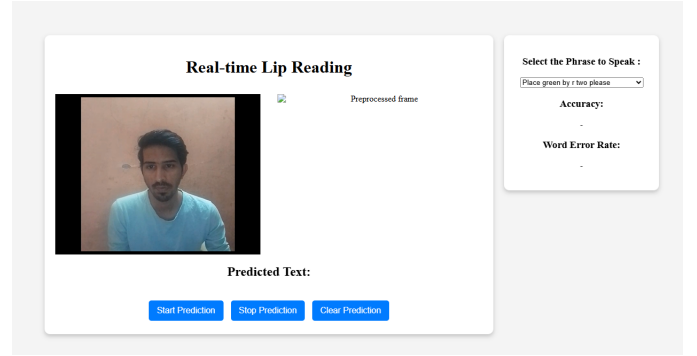


Fig. 5. UI Implementation for Real-Time Lip Reading

cess. This is done to maintain low-latency and near-immediate feedback.

The loop continues to execute continuously until the user decides to stop the prediction process through the user interface. When the stop message is received, the frontend stops sending frames to the backend so that any pending frames can be processed. The BiLSTM states, which maintain temporal relationships for improved sequence prediction, are at this point evidently reset. This is required since the model is stateful, and keeping previous states after a session may result in inconsistencies in subsequent predictions. Resetting each new session ensures that it begins with a new state, which does not suffer from interference caused by previous input.

This systematic but efficient approach allows real-time lip reading to function unhindered even in hardware with no native GPU support. By prudent management of frame queues, the use of multithreading, and the generation of a seamless stream

of data, the system controls inference speed and processing efficiency to facilitate real-time lip reading in a computing environment with limited capabilities. Frontend UI Implementation is depicted in Fig. 5.

This model demonstrated in Fig. 4 was tested on a real-time system to evaluate its performance in predicting phrases as spoken. The system was deployed as a live video processor via webcam, capturing frames at 30 FPS and providing them to the trained model in batches of 25 frames. Each batch's predictions were then combined to generate the final phrase for the whole video. This real-time setup was employed to evaluate the performance of the model in accurately transcribing lip movements. Despite the official training paradigm and model optimizations, the hoped-for results could not be achieved in real-time inference. The model was not able to generate significant predictions, and even after applying edit distance correction, the output remained predominantly inaccurate with a best-case accuracy of 50% for some phrases. The offline and real-time performance difference reveals deep problems with deploying lip-reading systems into real, dynamic settings. The problems encountered with real-time inference, as well as areas of improvement, are discussed in the following sections.

2) *Challenges Encountered in Real Time Execution:* Real-time deployment of the lip-reading model proved to have some issues as can be observed from Fig. 6 that may not exist in the offline scenario.

- **Latency:** While WebRTC minimized the delay in frame delivery, model inference was still plagued by a perceptible lag. Since the system only predicts when 25 frames are queued, any delay in processing after the system waiting time resulted in a delay in providing real-time feedback. The delay was even more apparent when consecutive predictions had to be processed sequentially.
- **Environmental Variability:** Although offline controlled training environments offer consistent lighting, changing camera angles, and limited occlusions, real environments offer changing lighting, changing camera angles, and sporadic occlusions. Unstable lighting sometimes changed lip region contrast, which made it difficult for the model to acquire discriminative features. Similarly, slight changes in camera position affected the observability of significant lip movements, decreasing prediction accuracy.
- **Speaker Adaptation:** The system was trained on a formal data set in the first place, but real-world use introduced natural variations in lip movement patterns, pronunciation, and speech rates. Even though training data from five speakers helped with generalization, the system continued to have poor performance with new users, indicating further dataset expansion and domain adaptation techniques.
- **Frame synchronization:** The Model works with constant 25-frame blocks, yet during real-time inference, the arriving frames don't necessarily stick to the order expected. Tiny variations in the frame arrival time or random loss of frames led to incorrect phrase segmentation, and prediction errors resulted.
- **Speaking Speed Mismatch:** Mismatch between training

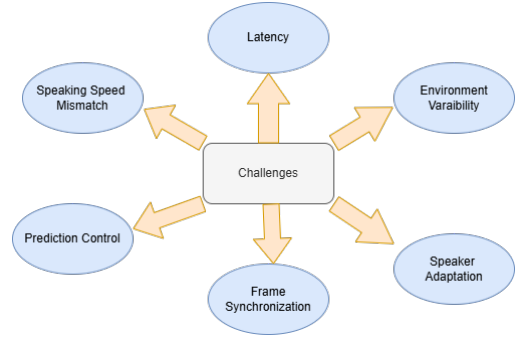


Fig. 6. Challenges encountered in Real time Lip Reading

data and actual scenarios regarding the speed of speaking. Speakers in GRID data speak at a uniform pace, whereas in actual scenarios, users speak either faster or slower than the dataset's average speed. Because the model anticipates lip movements to adhere to the patterns learned from training, speed of speaking differences can result in inaccurate predictions. The model might skip important frames when speech is faster, whereas slower speech might result in redundant repetition or incomplete sentence formation.

- **Prediction Control:** A challenge unique to this system was when a user requested the stop prediction action before the queue built up to 25 frames. Because the model was trained on sequences of precisely 25 frames, to predict anything on a sequence shorter than this led to unstable outputs. It was an issue because the model could not dynamically adjust to variable input lengths, leading to incomplete or incorrect phrase predictions when a premature stop was requested.

In addition, computational constraints were a challenge since the system was running without a GPU. Deep learning networks, especially those with 3D convolutions and recurrent networks, are typically optimized for hardware acceleration. CPU inference introduced processing bottlenecks, which, while mitigated by multithreading, contributed to system-level latency.

3) *Potential Improvements for Real-Time Lip Reading:* To improve the performance and stability of real-time lip reading, various modifications can be made in various components of the system. The prominent challenges such as latency, environmental variation, frame synchronization, and speaker adaptation need specific solutions to fill the gap between offline and real-time performance.

- Reducing latency and computation bottlenecks are among the most critical points of improvement. While hardware accelerators such as GPUs or special AI chips can speed up inference significantly, model processing time is the root cause of latency. Model quantization and pruning [14] methods can reduce the computational burden by lowering the precision of model parameters or pruning redundant weights to expedite inference without losing accuracy. Other optimizations such as model com-

pilation using TensorRT or executing it on edge-friendly frameworks can enhance real-time responsiveness.

- Frame synchronization problems are another vital enhancement. Because the model is operating on fixed 25-frame blocks, any live input synchronization discrepancy can lead to partial or wrong predictions. Frame interpolation methods [12] can be incorporated to make sure that in missing frames, a more fluid input stream is maintained.
- For handling variability in the environment, more pre-processing can be beneficial. Integration of real-time face tracking and lip area stabilization [13] can assist in minimizing inconsistencies due to camera motion and varying illumination. Adaptive contrast enhancement [11] can provide consistency in lip visibility across varying conditions.
- Speaker adaptation can be boosted through augmented training data and applying domain adaptation techniques [19]. Training on a larger, more heterogeneous set of speakers can make the model better generalize to unseen speakers. Fine-tuning on actual real-time user data of users conversing with the system can also make it more adaptable. Adaptive normalization techniques or few-shot learning can also make the model adapt better to different speaking styles.
- Another important improvement is handling cases where prediction is triggered before 25 frames are accumulated. Another strategy could be to modify the model to handle variable-length sequences so that it can make reasonable predictions even for incomplete sequences of less than 25 frames. This can be achieved by sequence padding, attention-based methods that learn to handle short sequences, or a fallback method where the system discards incomplete input rather than making suspicious predictions.
- To address the challenge the disparity of speaking pace in training and real conditions poses, Dynamic Time Warping (DTW) [16] or Adaptive Temporal Alignment techniques[17] can be explored. The methods allow the model to account for varying speaking paces by more adaptively aligning real-time input against learned patterns. Data training augmentation [18] for the model, synthetically speeding up or slowing down training videos to make the model encounter a wider range of speaking paces, is another potential method.
- Lastly, exploring other models such as Transformer-based models or hybrid models, ensemble learning, combining convolutional and self-supervised learning approaches may enhance robustness. Transformers, as capable of processing long-range dependencies, may offer enhanced phrase-level prediction, and self-supervised learning may enable the model to be trained on learning more about regular real-world differences without requiring large amounts of labeled data.

By integrating these improvements, the real-time lip-reading system can be made more accurate, more robust, and more re-

sponsive, closer to real-world deployment in real applications.

IV. DISCUSSIONS

The design of a system for real-time lip-reading is a complex set of issues. All bear on the performance and usability of the system. This presentation specifies the implications of the work by recognizing its constraints and outlining the avenues for further research.

A. Implications of this Research

The creation of an effective real-time lip-reading system has high potential in various applications. For the deaf, it could be an effective communication aid that translates visual speech signals into text or synthesized speech. In noisy environments where audio signals are masked, lip-reading technology could be used to enhance speech recognition systems, enhancing human-computer interaction. For security and surveillance purposes, the ability to read speech without hearing it can be used for the benefit of protecting privacy but can still be made operational. Real Time Lip-reading technology could enhance biometric security systems through the provision of an additional factor of authentication. The examination of the unique lip movement patterns over certain sentences could be used as a biometric signature, enhancing the security systems.

B. Limitations

Despite the advancements achieved, several limitations persist. The model's reliance on the GRID dataset, characterized by its structured and limited vocabulary, restricts the system's ability to generalize to more naturalistic and spontaneous speech patterns. The absence of a GPU in the deployment environment imposes computational constraints, leading to increased latency and reduced real-time performance. Additionally, the system's sensitivity to variations in lighting, speaker appearance, and speaking styles underscores the need for more robust preprocessing and adaptive learning mechanisms.

V. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we introduced and experimented with a deep learning-based real-time lip-reading model that could predict the spoken words from silent video input. We used a combination of a 3D CNN for spatial and temporal feature extraction, an attention mechanism to focus further on the salient features, and a BiLSTM backend for modeling sequences. The model also utilized Connectionist Temporal Classification(CTC) as a loss function. We trained the model on the GRID dataset, adopting a systematic training process with a fixed initial learning rate and a dynamic learning rate scheduler for enhanced convergence. Performance testing, by Word Error Rate (WER), demonstrated the efficiency of the model, with a WER of 2.81%, or an approximate accuracy of 97.2%.

While implementing this real-time system, there were several challenges, but most importantly the challenge of achieving efficiency in the processing speed without sacrificing on prediction accuracy. Real-time processing of inference without

overburdening the system with too much computational load was one of the most fundamental challenges. Application of stateful LSTMs with resetting the state after each video and not after each chunk was also one of the most fundamental optimizations for processing sequential dependencies in an efficient manner. Chunk processing of frames into 25 and strict pre-processing steps like cropping of the lip areas, conversion to grayscale, normalization, and resizing ensured input data were fed to the model in the most efficient manner possible.

We also present the problems encountered during the development of this real-time application and how they can be avoided. Despite the challenges, this work provides a significant contribution to real-time lip reading improvement, indicating the potential for deep learning-based approaches to silent speech understanding. As model structure, training algorithms, and optimization strategies continue to evolve, real-time lip-reading systems are potentially valuable tools in a range of applications from hearing-impaired accessibility through to security and silent communications technologies. Addressing these limitations offers a pathway to enhance the system's efficacy:

- **Large and Multifaceted Datasets:** Scaling training to large vocabularies and spontaneous speech can allow the model to generalize better to real-world situations.
- **Improved Hardware Utilization:** Using GPUs or hardware accelerators can reduce computational bottlenecks, making inference faster and real-time interaction more fluid.
- **Pre-processing Robustness Techniques:** Pre-processing techniques with adaptability, i.e., real-time tracking of faces, Data Augmentation can enhance system robustness to changes in the environment.
- **Speaker Adaptation Processes:** Domain adaptation and transfer learning processes can enable the model to acquire speaker-dependent features and hence improve performance on different groups of users.

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